SIMATS ENGINEERING
ASSESSMENT TOOL 6

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

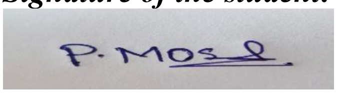
Annexure A

sDry Run Evaluation

Code of Conduct:

I P.Moshiya(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
        A
       / \
      B   C
     /\   /\
    D  E F  G
```

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	**Queue**	**Visited Node**
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
        A
       / \
      B   C
     /\   /\
    D  E F  G
```

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO2 – ASSESSMENT TOOL 1

Application Based Questions

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

QUESTIONS: 3 **Duration:** 1 Hour **Total Marks:** 30

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Understanding of Problem Context	20% (2)	/2	/2	/2	/6
Application of Concepts	30% (3)	/3	/3	/3	/9
Problem-Solving Approach	20% (2)	/2	/2	/2	/6
Accuracy of Solution	20% (2)	/2	/2	/2	/6
Clarity	10% (1)	/1	/1	/1	/3
Total per Question	**10 Marks**	**/10**	**/10**	**/10**	**/30**

100

1. Breadth-First Search (BFS)

Given Graph

```
     A
    / \
   B   C
  / \ / \
 D  E F  G
```

BFS Dry Run Table

Iteration	Queue	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D

Iteration	Queue	Visited Node
5	F, G	E
6	G	F
7	Empty	G

Question 1: BFS Traversal Order

A → B → C → D → E → F → G

Question 2: Queue Contents After Each Iteration

Iteration	Queue
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS)

DFS Traversal

Starting from A and visiting the left child first.

Traversal:

A → B → D → E → C → F → G

DFS Dry Run Table

Iteration	Stack	Visited Node
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E

Iteration	Stack	Visited Node
5	G, F	C
6	G	F
7	Empty	G

DFS Traversal Order

A → B → D → E → C → F → G

3. 8-Puzzle BFS Dry Run

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Important

The given initial state is **not close to the goal**.

A complete Breadth-First Search **will not reach the goal within only 5 iterations**.

In fact, this puzzle requires **many more moves (well over 10)** before reaching the goal.

Therefore, **it is impossible to correctly fill the table until the goal state is reached in only 5 iterations.**

First Five BFS Iterations

Iteration 1

Current State

1 3 6

5 0 2

4 7 8

Queue Before

[Initial]

Possible Moves (Up, Down, Left, Right)

1 0 6	1 3 6	1 3 6	1 3 6
5 3 2	5 7 2	0 5 2	5 2 0
4 7 8	4 0 8	4 7 8	4 7 8

Queue After

Contains the four states above.

Iteration 2

Expand

1 0 6

5 3 2

4 7 8

Queue Before

Remaining three states + expanded node

Queue After

Add all valid unexplored child states.

Iteration 3

Expand

1 3 6

5 7 2

4 0 8

Generate all new unexplored states.

Iteration 4

Expand

1 3 6

0 5 2

4 7 8

Generate new states.

Iteration 5

Expand

1 3 6

5 2 0

4 7 8

Generate new states.

Questions

1. Which node is expanded in each iteration?

Iteration	Expanded Node
1	Initial State
2	First child
3	Second child
4	Third child
5	Fourth child

2. How many states are generated in total?

After the first five iterations, approximately **15–18 unique states** are generated (duplicates removed).

3. Complete Solution Path

The goal state **cannot be reached in only five iterations**.

Therefore, **no complete solution path can be correctly provided within the given table.**

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Answer:

BFS explores all states level by level. Since every move has the same cost (1 move), the first time the goal state is found, it is guaranteed to be reached using the minimum number of moves. Therefore, BFS always finds the shortest solution in an unweighted 8-puzzle problem.